

Christian Lee John, Ph.D.

Assistant Professor
Michigan Technological University

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Education and training

University of California, Santa Barbara – Postdoctoral scholar (2022 –2026)
University of California, Davis – Ph.D. Ecology (2022)
The Pennsylvania State University – M.Sc. Ecology (2016)
The Pennsylvania State University – B.S. Biology (2013)

Teaching

Instructor

Environmental Data Science (EDS) Capstone (MTU)	Fall 2026
<i>Upcoming: EDS Capstone (MTU)</i>	<i>Spring 2027</i>
<i>Upcoming: Modeling Species-Environment Relationships (MTU)</i>	<i>Spring 2027</i>

Graduate student co-instructor

R-DAVIS (UC Davis): Intro to Data Analysis and VISualization; lab	Fall 2020
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Teaching Assistant

R-DAVIS (UC Davis): Intro to Data Analysis and VISualization; lab	Winter 2019, 2020
ENT 117 (UC Davis): Longevity; lecture	Summer 2018
PLS 147 (UC Davis): California Plant Communities; lecture	Spring 2018
WFC 198 (UC Davis): Climate Change Ecology; discussion	Winter 2018
BIOL 415 (Penn State): Ecotoxicology; lecture	Spring 2014, 2017
BIOL 220 (Penn State): Population and Community Biology; lab	Spring 2015, 2017
BIOL 110 (Penn State): Basic Concepts and Biodiversity; lab	Summer, Fall 2013

Peer-Reviewed Publications

24. **John, C.**, O'Connor, B., Stephenson, T.R., and Post, E. 2026. Landscape Greening Following Unseasonal Precipitation Along a Desert–Alpine Gradient. *Remote Sensing*. 18(17) 2951. [\[article\]](#)
23. Speare, K.E. and 23 others (incl. **John, C.**). 2026. Reducing consumer pressure increases community dissimilarity leading to coral- or algal-dominance on a coral reef. *Ecology*. [\[article\]](#)
22. Pick, J.L. and 26 others (incl. **John, C.**). 2026. The SORTEE Guidelines for Data and Code Quality Control in Ecology and Evolutionary Biology. *Peer Community Journal*. [\[article\]](#)
21. **John, C.** and 28 others. 2025. Terrigenous inputs link nutrients regimes to microbial communities in a tropical lagoon. *Limnology and Oceanography*. [\[article\]](#) [\[press release\]](#)
20. Neumann, K.*, **John, C.***, Atger, T., Punu, T., Hollarsmith, J.A., and Burkepile, D.E. 2025. Land use shapes riverine nutrient and sediment concentrations on Moorea, French Polynesia. *Scientific Reports*. 15:27948. (*authors contributed equally to this work) [\[article\]](#)

19. Edmunds, P., **John, C.**, Leichter, J., Moritz, C., Scafidi, K., Speare, K., Srednick, G., and Wyatt, A., 2025. Long-term community dynamics are heterogeneous among habitats on an Indo-Pacific coral reef. *Ecosphere*. [[article](#)]
18. Post, E., Bøving P.S., Higgins, R.C., **John, C.**, Post, M., O'Connor, B., Reimer, J., Williamson, N., and Kerby, J.T. 2025. A framework for reproductive outcomes of phenological match-mismatch in migratory breeders tested on a declining species, caribou. *PNAS Nexus*. 4(6): pgaf188. [[article](#)]
17. Detmer, A.R., Miller, S.D., Ring, K., **John, C.**, Briggs, C., Rassweiler, A., and Moeller, H. 2025. A Busse balloon in the lagoon: herbivore behavior generates spatial patterns in coral reef ecosystems. *Ecology Letters*. 28(3) 1-13. [[article](#)]
16. **John, C.**, Avgar, T., Rittger, K., Smith, J.A., Stephenson, L.W., Stephenson, T.R., and Post, E. 2024. Pursuit and escape drive fine-scale movement variation during migration in a temperate alpine ungulate. *Scientific Reports*. 14:15068. [[article](#)]
15. Post, E., Higgins, R.C., Bøving P.S., **John, C.**, Post, M., and Kerby, J.T. 2024. Large herbivores link plant phenology and abundance in arctic tundra. *PNAS Nexus*. 3(11):pgae454. [[article](#)]
14. **John, C.**, Kerby, J., Stephenson, T.R., and Post, E. 2023. Fine-scale landscape phenology revealed through time-lapse imagery: implications for conservation and management of an endangered migratory herbivore. *Remote Sensing in Ecology and Conservation*. 9(5) 628-640. [[article](#)]
13. Post, E., and 11 others (incl. **John, C.**). 2023. Large herbivore diversity slows sea ice-associated decline in arctic tundra diversity. *Science*. 380:6651. [[article](#)]
12. **John, C.** and Post, E. 2022. Seasonality, niche management, and vertical migration in landscapes of relief. *Ecography*. 2022(6):E05774. [[article](#)]
11. Strong, D. and **John, C.** 2022. Moisture and Distribution of a Keratophagous Moth, *Tinea occidentella*. *Ecological Entomology*. 47(5) 822-830. [[abstract](#); contact for article]
10. **John, C.** and Post, E. 2022. Projected distribution of Nearctic Bovidae signals the potential for reduced overlap with protected area. *Ecology and Evolution*. 12(8):e9189. [[article](#)]
9. Berger, D., German, D.W., **John, C.**, Hart, R., Stephenson, T.R., and Avgar, T. 2022. Seeing is be- Leaving: Perception informs migratory decisions in Sierra Nevada Bighorn Sheep (*Ovis canadensis sierrae*). *Frontiers in Ecology and Evolution*. 10:742275. [[article](#)]
8. Vuorinen, K. and 41 others (incl. **John, C.**) 2022. Growth rings show limited evidence for ungulates' potential to suppress shrubs across the Arctic. *Environmental Research Letters*. 17(3) 034013. [[article](#)]
7. **John, C.**, Shilling, S., and Post, E. 2021. drpToolkit: An automated workflow for aligning and analyzing vegetation and ground surface time series imagery. *Methods in Ecology and Evolution*. 13(1) 54-59. [[article](#)]
6. Eikelenboom, M., Higgins, R.C., **John, C.**, Kerby, J., Forchhammer, M.C., and Post, E. 2021. Contrasting dynamical responses of sympatric caribou and muskoxen to winter weather and earlier spring green-up in the Arctic. *Food Webs*. 27 e00196. [[abstract](#); contact for article]
5. Myers-Smith, I.H., Kerby, J.T., Phoenix, G.K., Bjerke, J.W., Epstein, H.E., Assmann, J.A., **John, C.**, et al. 2020. Complexity revealed in the greening of the Arctic. *Nature Climate Change*. 10(2) 106-117. [[article](#)]
4. **John, C.** Miller, D., and Post, E. 2020. Regional variation in green-up timing along a caribou migratory corridor: Spatial associations with snowmelt and temperature. *Arctic, Antarctic, and Alpine Research*. 52(1) 416-423. [[article](#)]

3. Post, E., Beyen, E., Bøving, P., Higgins, R.C., **John, C.**, Kerby, J., Pedersen, C., and Watts, D. 2019. Unusual late July observation of a fledgling Lapland longspur in low arctic Greenland following the late spring of 2018. *Arctic Science*. 5(3) 161-166. [[article](#)]
2. Blazanin, N., Son, J., Craig-Lucas, A., **John, C.**, Breech, K., Son, J., Podolsky, M., and Glick, A.B. 2017. ER Stress and Distinct Outputs of the IRE1 α RNase Control Proliferation and Senescence in Response to Oncogenic Ras. *Proceedings of the National Academy of Sciences*. 114(37) 9900-9905. [[article](#)]
1. Van den Bogaard, E.H., Podolsky, M.A., Smits, J.P., Cui, X., **John, C.**, et al. 2015. Genetic and Pharmacological Analysis Identifies a Physiological Role for the AHR in Epidermal Differentiation. *Journal of Investigative Dermatology*. 135(5) 1320-28. [[article](#)]

Other Publications

Book chapters

- John, C.** 2025. North American tundras: Imperiled landscapes at a continent's latitudinal and altitudinal extremes. In *Global Biome Conservation and Global Warming: Impacts on Ecology and Biodiversity*. G. L. Demolin-Leite Sr., P. Török, and J. C. B. Barrionuevo Sr. Editors. Elsevier. [[chapter](#)]

Technical reports and recommendations

- Pick, J.L. and 26 others (incl. **John, C.**). 2025. The SORTEE Guidelines for Data and Code Quality Control in Ecology and Evolutionary Biology. *EcoEvoRxiv*. [[guidelines](#)]
- Mejía-Ramón, A.G. and **John, C.** 2017. Informe Preliminar sobre las Actividades de Teledetección Aérea en la Zona Monumental Arqueológica de Teotihuacan, Verano 2017. Technical report submitted to the Consejo de Arqueología, October 2017, Instituto Nacional de Antropología e Historia, México D.F.

Theses and dissertations

- John, C.** 2022. Multi-scale spatial ecology of an endangered alpine migrant, Sierra Nevada bighorn sheep. Ph.D. Dissertation, University of California, Davis. [[dissertation](#)]
- John, C.** 2016. Against the Spring Wave: Ungulate Migration Phenology in a Changing Arctic. M.Sc. Thesis, Penn State University. [[thesis](#)]

Software

- John, C.** 2023. phenomap: Projecting satellite-derived phenology in space. R package version 2.0.1. <<https://CRAN.R-project.org/package=phenomap>>.
- John, C.** Shilling, F., and Post, E. 2021. drpToolkit: Digital repeat photography imagery management and analysis. Python package version 1.0.0. <<https://github.com/JepsonNomad/drpToolkit>>.
- John, C.** Bolas, E., and Laca, E. 2020. ggeAdmit: Streamlining the holistic review process. R package version 0.0.1. <<https://github.com/ggeDCAA/ggeAdmit>>.

Presentations

Listed in order of recency. IO = Invited oral presentation; CO = Contributed oral presentation;
CL = Contributed lightning talk; P = Poster presentation

Ifremer, Université de la Polynésie Française [IO] Sep 2025
John, C. *Land use, precipitation, and nutrient variability in waters of Moorea*

University of California system-wide GIS Week [CO] Nov 2024
John, C. *Nearshore nutrient regimes link land use to tropical lagoon microbial communities*

UC Davis Department of Wildlife, Fish, and Conservation Biology [IO]	Oct 2024
John, C. <i>Ridge-to-reef connections and community response to seasonal stress in a tropical lagoon system</i>	
Ecological Society of America Summer 2024 Meeting [CO]	Aug 2024
John, C. , Speare, K., Adam, T., Donovan, M., and Burkepile, D. <i>Consistent thermal regimes drive repeated spatial patterning in stress-induced mortality of foundation species</i>	
Sierra Nevada Science Symposium [CL]	Mar 2024
John, C. , Stephenson, T., and Post, E. <i>Time-lapse cameras reveal fine-scale patterns of plant phenology in the eastern Sierra</i>	
Ecological Society of America Summer 2023 Meeting [CO]	Aug 2023
John, C. , Neumann, K., Adam, T., Donovan, M., and Burkepile, D. <i>Land use change, nutrient pollution, and sedimentation seasonality in a South Pacific coral reef island system</i>	
UC Santa Barbara Department of Ecology, Evolution, and Marine Biology [IO]	Oct 2022
John, C. <i>Seasonality and predictability of community response to anthropogenic change</i>	
Aarhus Universitet Department of Biology [IO]	Mar 2022
John, C. <i>Linking food on the ground to a satellite pixel: Phenological insights from a time-lapse camera network</i>	
American Geophysical Union Fall 2021 Meeting [P]	Dec 2021
John, C. , Stephenson, T.R., and Post, E. <i>Detecting Plant Phenological Response to Snowmelt on “Barren” Alpine Landscapes</i>	
American Geophysical Union Fall 2020 Meeting [P]	Dec 2020
John, C. , Stephenson, T., and Post, E. <i>Associations among topography, snowmelt, and plant green-up phenology, revealed through a time-lapse camera network</i>	
UC Davis Polar Forum Mini Symposium [CO]	Nov 2019
John, C. and Post, E. <i>Spatial disparity in drivers of landscape phenology regulates regional green-up dynamics</i>	
American Geophysical Union Fall 2018 Meeting [CO]	Dec 2018
John, C. , Stephenson, T., and Post, E. <i>Long-term projections for migration phenology in Sierra Nevada bighorn sheep</i>	
Ecological Society of America Summer 2017 Meeting [CO]	Aug 2017
John, C. and Post, E. <i>Arctic green wave dynamicity and its implications for migration phenology</i>	
American Geophysical Union Fall 2016 Meeting [CO]	Dec 2016
John, C. and Post, E. <i>Dissecting drivers of Arctic plant phenology across scales in time and space</i>	
GIS Day at Penn State, State College, PA [CL]	Nov 2015
John, C. <i>Using UAV remote sensing in Arctic ecological research</i>	
The Polar Center at Penn State, State College, PA [CO]	Feb 2015
John, C. and Post, E. <i>Climate change, the green wave, and Arctic ungulate migratory timing</i>	

Peer Reviews

Peer reviews provided for manuscripts in the following publications:

The American Naturalist; Animal Migration; Arctic, Antarctic, and Alpine Research; Biological Conservation; Canadian Journal of Zoology; CLEAN Soil Air Water; Current Biology; Ecography; Ecological Applications; Ecology; Ecology and Evolution;

Ecology Letters; Environmental Pollution; Global Change Biology; Landscape Ecology; Oikos; Remote Sensing in Ecology and Conservation; Scientific Data; Scientific Reports; Water Research X

Ad-hoc review provided for research proposal in:
NSF Arctic System Science (ARCSS) Program 2024

Other Professional Service

SORTEE Treasurer and Interim Member, Board of Directors	2026
Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology; managing society finances, and contributing to operations and programming.	
SORTEE Advocacy Committee member	2025-2026
Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology; contributing to Advocacy Committee to advance reproducible and open-access scientific best practices.	
Mapping Bonanza: Organizer and instructor	Spring 2024
Developed a 4-part workshop miniseries to introduce spatial data wrangling, analysis, and visualization for graduate students and postdocs (UC Santa Barbara).	
Statistical Support Group co-coordinator	Spring 2022
Providing advice and leading discussion about experimental design, statistics, and related issues in an informal weekly graduate student meet-up (UC Davis).	
Admissions Planning Committee student representative	Fall 2021 – 2022
Implementing holistic review for graduate admissions in the UC Davis Graduate Group in Ecology.	
Admissions and Awards Subcommittee student member	Spring 2019 – 2022
Analyses of utility and effects of holistic review on graduate admissions; developed admissions reviewer assignment software; organized online-format admissions and reviewer training for remote learning during COVID-19 pandemic. Formally, graduate student researcher in Fall 2019.	
Open Lab Meeting co-coordinator	Fall 2020 – 2022
Graduate student-led “lab meeting” of weekly seminars within the graduate group community. No professors, relaxed atmosphere for candid presentation feedback from peers.	
Davis R Users’ Group co-coordinator	Fall 2020 – 2022
Weekly working group of R coding enthusiasts across undergraduate, graduate, and post-doc levels of training. Troubleshooting, workshops, and R-related discussion.	
Guest Lecturer and Other University Speaking Forums	
WFC 168 (UC Davis): <i>Snow, plant, and wild sheep phenol. in CA mtns</i>	Feb 2022
WFC 168 (UC Davis): <i>The future of phenology for Sierra bighorn</i>	Feb 2021
#MapTimeDavis (UC Davis) mapping workshop [link]	Jan 2021
WFC 010 (UC Davis): <i>Movement ecology</i>	Mar 2019
WFC 198 (UC Davis): <i>Resources, reproduction, and trophic mismatch</i>	Feb 2019
WFC 010 (UC Davis): <i>Animal migration</i>	Nov 2017, Feb 2020
WFS 585 (Penn State): <i>Phenoscape: Analyzing the green wave in R</i>	Mar 2017
GEOG 455 (Penn State): <i>Remote sensing in migration ecology</i>	Apr 2016